

Operating System

Practice Question (06-05-2024)

- Considering a system with five processes P_0 through P_4 and three resources types A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time t_0 following snapshot of the system has been taken:

Process	Maximum Need			Allocation		
	A	B	C	A	B	C
P0	7	5	3	0	1	0
P1	3	2	2	2	0	0
P2	9	0	2	3	0	2
P3	2	2	2	2	1	1
P4	4	3	3	0	0	2

- Find the Need Matrix?
 - Find whether system is in Deadlock or not? Write the safe state (if Possible)
 - What will happen if process P_1 requests one additional instance of resource type A and two instances of resource type C? Is additional request by process P_1 granted?
- Draw the resource allocation graph and determine deadlock or deadlock free state.

$$P = \{P_1, P_2, P_3\}$$

$$R = \{R_1, R_2, R_3, R_4\}$$

$$E = \{P_1 \rightarrow R_1, P_2 \rightarrow R_3, R_1 \rightarrow P_2, R_2 \rightarrow P_2, R_2 \rightarrow P_1, R_3 \rightarrow P_3, P_3 \rightarrow R_2\}$$

There are ONE, TWO, ONE and THREE instances of R_1, R_2, R_3 and R_4 type resources

- Consider the following set of processes, with length of CPU bursts given in Millisecond as follows:

Process	Burst Time	Arrival Time	Priority
P1	8	0	3
P2	1	1	1
P3	3	2	2
P4	2	3	3
P5	6	4	4

- Draw the Gantt Charts for FCFS, SJF, Preemptive priority and RR(Quantum=2)

- b. What is the turnaround time of each process for above algorithm?
 - c. What is the waiting time of each process for each of the above algorithm?
 - d. Which algorithm results in minimum average waiting time?
4. Suppose we have a paging system with a physical memory capacity of 4 frames and a virtual memory with a capacity of 8 pages. We're given the following page reference string: 1, 2, 3, 4, 5, 6, 1, 3, 7, 8, 1, 2, 3, 4, 5, 6. Find Page Fault with FIFO, LRU and Optimal Page Replacement Algorithm.
 5. Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is _____.
 6. Explain Free Space management in Secondary Memory.
 7. What is various file allocation methods. Explain Index File Allocation Method
 8. What are various Directory Structures. Explain them in Detail.
 9. Given a virtual memory system with a page size of 4KB and a 32-bit address space: a. How many bits are used for the page number and offset? b. Calculate the logical address when the page number is 256 and the offset is 1024.
 10. In a virtual memory system with a physical memory size of 16GB and a page size of 4KB: a. How many pages are there in the virtual address space? b. If each frame in physical memory is 4KB, how many frames are there? c. What is the size of the page table assuming each entry is 4 bytes?
 11. Given a virtual memory system with a 48-bit virtual address space and a page size of 8KB: a. How many bits are used for the page number and offset? b. Calculate the size of the page table assuming each entry is 4 bytes.
 12. Consider a disk with 100 cylinders (numbered from 0 to 99) and the disk scheduling algorithm SCAN, C-SCAN, Look, and C-Look. The disk arm is currently at cylinder 50 and moving towards cylinder 0. The following requests, in cylinder order, are to be serviced: 82, 20, 35, 67, 90, 42, 15, 70.